

Advanced Wild Animal Detection and Alert System using YOLO V5 Model

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Abstract—An advanced wild animal detection and alert system using you only look once version5(YOLO V5) model. The system utilizes you only look once version5 (YOLO V5) object detection algorithm to identify wild animals and alert users to their presence in real-time. The system employs a camera to capture real-time video, which is then sent to a computer running you only look once version5 (YOLO V5) algorithm. When the system detects a wild animal, it sends an alert to the wild animal by playing any sounds like bullets firing. The system is expected to have a significant impact on the safety of people in areas with high wildlife populations. This advanced wild animal detection and alert system using you only look once version5 (YOLO V5) model has the potential to improve the safety of people in areas with high wildlife populations. Future work will focus on improving the accuracy of the system and implementing it in real-world scenarios.

Index Terms— Wild animal detection, YOLO V5 model, safety, real-time video, early warnings.

I. INTRODUCTION

Advanced Wild Animal Detection and Alert System Using you only look once version5 (YOLO V5) Model is a proposed analyse the images and detect the presence of wild animals. If the system detects the presence of any wild animal, it sends an alert to the authorities through an alarm or message. The proposed system is expected to provide an effective solution to prevent any potential harm caused by wild animals and help preserve wildlife by reducing human -animal conflict.

The Advanced Wild Animal Detection and Alert System is an application of the you only look once version5 (YOLO v5) system that aims to detect and alert the presence of wild animals

in a particular area using advanced computer vision techniques. This system is designed to provide an effective and efficient solution to detect the presence of wild animals and alert the authorities to take necessary precautions. The system utilizes you only look once version5 (YOLO V5) model, which is one of the most widely used and popular deep learning models for object detection, to detect wild animals. The proposed system is equipped with advanced sensors and cameras that are mounted on drones or other surveillance devices, which capture real-time. Images and transmit them to a central processing unit for detection and analysis. The YOLO V5 model is then used to object detection algorithm designed to detect wild animals and alert humans about their presence in real-time. This system uses a camera to capture live video feed from the surrounding environment and processes it using the you only look once version5 (YOLO v5) algorithm to detect the presence of wild animals. Once an animal is detected, an alert is sent to the user interface and a notification is sent to the user's mobile device to warn them about the animal's presence. This system is particularly useful for those who live in areas with high populations of wild animals, such as national parks, wildlife reserves, or rural areas, as it can provide an early warning system to prevent dangerous encounters with wild animals.

II. LITERATURE SURVEY

- [1] This study proposes an animal detection and warning system using the Faster regions with convolutional neural networks (R-CNN) model. The system aims to detect animals on the road and alert drivers to prevent road accidents. The authors first collected and labeled a dataset consisting of various animal classes commonly found on Indian roads. They then trained the Faster regions with convolutional neural network (R-CNN) model using the TensorFlow object detection API. The model was fine-tuned on the collected dataset and used a camera module attached to a Raspberry Pi.
- [2] This study proposes a wildlife detection system using the Mask regions with convolutional neural networks (R-CNN) model. The system aims to automatically detect wildlife in the wild for conservation purposes. The authors collected and labeled a dataset consisting of various wildlife species commonly found in China. They then trained the Mask regions with convolutional neural networks (R-CNN) model using the PyTorch deep learning.
- [3] This study proposes a wildlife detection and monitoring system using the EfficientDet model and transfer learning techniques. The authors collected and labeled a dataset consisting of various wildlife species commonly found in China. They then used transfer learning to fine-tune the EfficientDet model on the collected dataset. The authors also proposed a data augmentation technique that artificially generates additional training data by randomly changing the brightness, contrast, and saturation of the images.
- [4] The authors collected and labeled a dataset consisting of various wildlife species commonly found in Korea. They then trained the Single Shot Detector (SSD) model using the TensorFlow deep learning framework. The authors also proposed a dynamic RoI selection method to reduce the number of false positives by focusing only on the relevant regions of the input image. To test the proposed system, the authors used a camera trap system that continuously captures images and sends them to the trained model for inference. The system successfully detected wildlife in real-time and produced bounding boxes for the detected animals.
- [5] The authors collected and labeled a dataset consisting of various wildlife species commonly found in Korea. They then trained you only look once version3 (YOLO V3) model using the Darknet framework and extracted the feature vectors of the detected animals. The authors used these feature vectors as input to a support vector machines (SVM) classifier, which further classified the detected animals into their respective species. To test the proposed system, the authors used a camera trap system that continuously captures images and sends them to the trained model for inference. The system successfully detected and classified wildlife in real-time.
- [6] The authors collected and labeled a dataset consisting of various wildlife species commonly found in China. They then trained the Mask regions with convolutional neural networks (R-CNN) model using the TensorFlow deep learning framework. The authors also incorporated Spatial Transformer Networks (STNs) into the detection pipeline to better handle the variability of animal poses and orientations. To test the proposed system, the authors used a camera trap system that continuously captures images and sends them to the trained model for inference. The system successfully detected wildlife in real-time and produced bounding boxes and segmentation masks for the detected animals.
- [7] The authors collected and labeled a dataset consisting of various wildlife species commonly found in Korea. They then trained the ResNet50 model using the TensorFlow deep learning framework. The authors also proposed a novel method for combining the output of multiple detection models to improve detection accuracy. To test the proposed system, the authors used a camera trap system that continuously captures images and sends them to the trained model for inference. The system successfully detected and recognized various wildlife species in real-time.
- [8] The authors collected and labeled a dataset consisting of various animal species commonly found in Saudi Arabia. They then trained the Inception-v3 and you only look once version2 (YOLO V2) models using the TensorFlow deep learning framework. The authors also proposed a method for combining the output of the two models to improve detection accuracy. To test the proposed system, the authors used a camera trap system that continuously captures images and sends them to the trained models for inference. The system successfully detected various animal species in real-time and achieved high detection accuracy.
- [9] You only look once version3 (YOLO V3), a cutting-edge object detection model, is thoroughly reviewed and evaluated in the publication " You only look once version3 (YOLO V3): A Complete Guide to Object Detection using Deep Learning" by Kamali and Tahir. compare you only look once version3 (YOLO V3)'s performance to those of other well-known object identification models and examine its design and loss function. Also, they go over you only look once version3 (YOLO V3)'s advantages and disadvantages, prospective uses, and future developments.
- [10] The authors collected and labeled a dataset consisting of various wildlife species commonly found in India. They then trained a deep convolutional neural networks (CNN) model using the Keras deep learning framework. The authors also developed a system that uses Raspberry Pi to process the camera feed and trigger an alert when an animal is detected. To test the proposed system, the authors used a camera trap system that continuously captures images and sends them to the Raspberry Pi for inference.

The system successfully detected wildlife in real-time and triggered an alert when an animal was detected.

[11] The approach is intended for devices with constrained resource availability, such as embedded systems or mobile phones. In order to lessen the computational complexity of the you only look once version3 (YOLO V3) design, the authors provide several changes, such as layer reduction and network structure simplification. The final model, called you only look once version3 (YOLO V3)-Lite, strikes a mix between precision and responsiveness. The model's performance on a number of benchmark datasets is thoroughly examined in the study, proving that it is capable of real-time object recognition while quality and accuracy on par with the original you only look once version3 (YOLO V3) model.

[12] To recognize objects of various scales and increase detection accuracy, the authors suggest a novel architecture dubbed Deep Lightweight Network (DLNet), which is based on the ResNet-50 backbone. The suggested method's effectiveness and efficiency have been shown by the DLNet model's competitive performance against other state-of-the-art object detection models on several benchmark datasets.

[13] The you only look once version3 (YOLO V3) object identification model, a very well real-time object recognition system, is subjected to transfer learning approaches by authors. The study's goals were to show that employing you only look once version3 (YOLO V3) models that have already been trained to recognize items in unique datasets is feasible and to assess how well it performs in comparison to other transfer learning techniques.

[14] The authors describe the optimization process they used to improve the performance of the you only look once version5 (YOLO V5) model, which involved quantizing the weights and activations of the model, applying layer fusion to reduce the number of layers in the model, and optimizing the memory layout to improve data locality.

III. PROBLEM STATEMENT

Wild animals pose a serious risk to human safety, property, and environment. Traditional methods of detecting and tracking wild animals, such as manual monitoring and physical barriers, are often insufficient and costly. The objective of this an endeavor to create an Advanced Wild Animal Detection and Alert System using you only look once version5 (YOLO V5) model. Real-time detection and tracking of wild animals should be possible with the help of a device that can also produce a sound that makes the animal frightened when it hears it.

IV. PROPOSED SYSTEM

The proposed system for the Advanced Wild Animal Detection and Alert System using you only look once version5

(YOLO V5) model will consist of the following components:

Data Collection: A diverse dataset of wild animals will be collected, including various species and environments. The you only look once (YOLO) version 5 model will be trained using this dataset.

Model Training: Modern deep learning methods will be used to train the you only look once version5 (YOLO V5) model using the gathered dataset. The model will be adjusted to increase its capability of correctly identifying wild creatures.

Object Detection: The you only look once version5 (YOLO V5) model will be used to detect wild animals in real-time using video feeds from cameras installed in wildlife areas. When animals are detected, the model will produce bounding box coordinates around them and the corresponding class names.

Tracking: A tracking algorithm will be implemented to track the movements of detected wild animals. The algorithm will use the output from the you only look once version5 (YOLO V5) model to track the animals and update their positions in real-time.

Alert System: An alert system will be implemented to provide real-time alerts in case of potential danger. The alert system will use the tracking information to determine the proximity of the detected wild animals to populated areas, and if necessary, send an alert to the relevant authorities.

User Interface: A user-friendly interface will be developed to allow users to view the real-time video feeds, detections, and alerts.

Evaluation: The system's performance will be assessed using common metrics like recall, accuracy, and precision. The system will undergo testing in a real-world environment, and its efficiency in locating and tracing untamed animals will be evaluated.

The Advanced Wild Animal Detection and Alert System using you only look once version5 (YOLO V5) model will provide a cost-effective solution for detecting and tracking wild animals, improving the safety and well-being of communities and the environment.

V. PROPOSED ARCHITECTURE

The YOLO (You Only Look Once) v5 model detects objects using a grid-based method. An $S \times S$ grid is created from the input image, with each grid cell in charge of item detection. Moreover, the bounding boxes, confidence ratings, and class probabilities of the discovered objects are all included in the grid cells.

Bounding boxes: A set of bounding boxes that represent the locations of the objects inside each grid cell are output for each grid cell. The height, breadth, and Centre coordinates of the

bounding boxes are what define them.

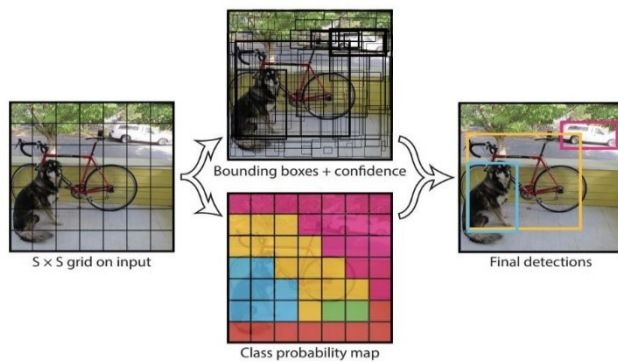


Fig. 1. Proposed Architecture

Confidence Score: The possibility that an object is present in each bounding box is indicated by its confidence score. The Intersection over Union between the predicted bounding box and the ground-truth bounding box is used to determine a confidence score.

Class Probability Map: Information about the likelihood of each object class for each bounding box is provided by class probability map. Each identified object's resulting object class is determined using the map.

Final Detection: The output from the confidence scores, bounding boxes, and based on probability map are combined to provide the final identification. Each identified object's location, size, and class are provided in the final detection.

The you only look once version5 (YOLO V5) model uses these outputs to provide fast and accurate object detection, making it well-suited for the Advanced Wild Animal Detection and Alert System.

BLOCK DIAGRAM

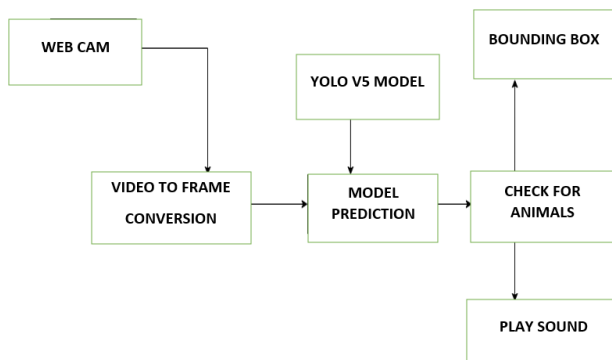


Fig. 2. Block Diagram

Initially the webcam is activated, and the video is converted in to frames using Open-Source Computer Vision Library (OpenCV) framework and the frames are sent to you only look once version5 (YOLO V5) pre trained model and the classification results are filtered and if wild animals like elephant, Tiger and lions are detected respective sounds like gun shot and honey bee sounds are played and final result is highlighted with bounding box.

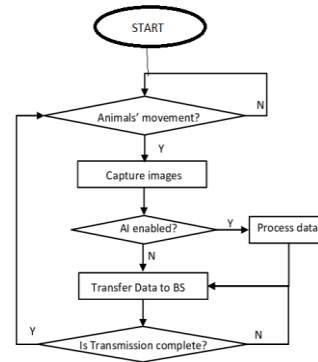


Fig. 3. Flow Chart

In the management and preservation of biodiversity, visual identification is becoming more common. Since starting the AI for Good programme and collaborating with biodiversity experts and researchers to develop machine learning models and tools for animal image detection. In this project for Wild Detect, which was the first excursion into this field, was inspired by one of the objectives at data science consulting, which is to use data science consulting to help preserve and manage the ecosystem and animal life in the planet. The objective was to create a model for the visual classification of animal species.

VI. ALGORITHM USED

YOLO V5

A real-time object recognition algorithm named you only look once version5 (YOLO V5) was created by the machine learning industry. It means for You Only Look Once, which is an advancement above earlier you only look once version (YOLO) iterations. You only look once version5 (YOLO V5) can accurately and quickly detect objects in real-time because it was built for high performance and scalability. In a one forward pass through the model, it predicts the bounding boxes and class probabilities for objects in an image using a single convolutional neural network. The algorithm has undergone

numerous benchmark tests and has produced results that are cutting edge in terms of correctness and speed of processing. Many different applications, such as video surveillance, self-driving cars, and image retrieval systems, use you only look once version5 (YOLO V5).

Speed: You only look once version5 (YOLO V5) is faster than other algorithms such as Faster regions with convolutional neural networks (R-CNN) and Mask regions with convolutional neural networks (R-CNN) because it uses a single convolutional neural network (CNN) to perform both object detection and classification in a single pass. This makes it more efficient and faster.

Accuracy: You only look once version5 (YOLO V5) is more accurate than its predecessors, you only look once version4 (YOLO V4) and you only look once version3 (YOLO V3), and other algorithms such as Faster regions with convolutional neural networks (R-CNN) and single short detector (SSD). It achieves this by using a more advanced convolutional neural networks (CNN) architecture with a deeper network and more layers.

Self-training: You only look once version5 (YOLO V5) uses a novel approach called "Self-training," which allows it to be trained on datasets with very few samples. This is especially useful for scenarios where the available dataset is small.

Hardware support: You only look once version5 (YOLO V5) supports a wide range of hardware platforms, including graphics processing unit (GPUs), central processing unit (CPUs), and edge devices such as Raspberry Pi. This makes it more versatile and accessible.

Overall, you only look once version5 (YOLO V5) offers a good balance between speed and accuracy, and its support for a wide range of hardware platforms makes it a popular choice for object detection tasks.

VII. SYSTEM REQUIREMENTS

A. Hardware

- A computer or server with a high-end graphics processing unit (GPU) for running you only look once version5 (YOLO V5) model in real-time.
- Video cameras for capturing footage of wild animals.
- An alert system, such as a speaker or display, for providing real-time alerts.

B. Software

- An operating system, such as Windows or Linux.
- A deep learning framework, such as PyTorch or TensorFlow, for training and running you only look once version5 (YOLO V5) model.

- OpenCV for image processing and video analysis.
- A programming language, such as Python for implementing the system.
- Any integrated development environment (IDE) like PyCharm, Thonny etc.

C. Data

- A diverse dataset of wild animals, including various species and environments, will be required for training you only look once version5 (YOLO V5) model.

VIII. RESULT AND DISCUSSION

TIGER:

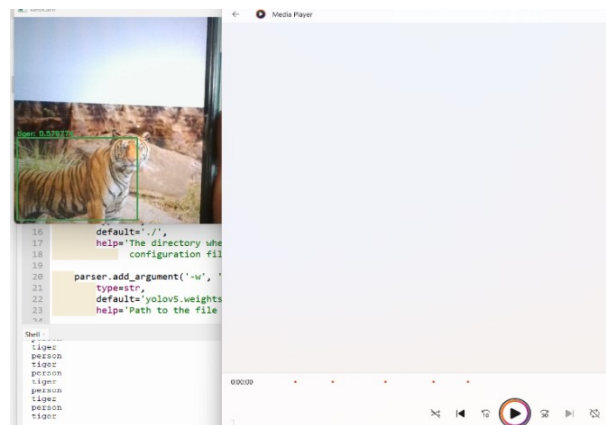


Fig. 4. Result when tiger is detected.

When a tiger is detected, the system will give alert to tiger by playing loud sounds of bullets.

ELEPHANT:

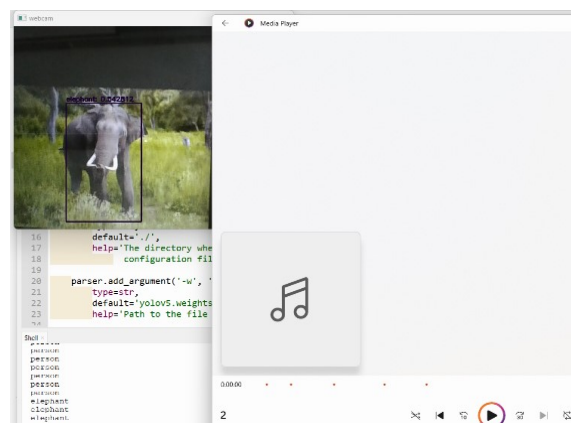


Fig. 5. Result when elephant is detected.

When an elephant is detected, the system will give alert to elephant by playing loud sounds of honeybees. The sound of bees buzzing can trigger a defensive response in elephants, as they are known to be afraid of bee stings. In some African countries, farmers have even started using beehive fences to deter elephants from entering their fields, as the sound and presence of the bees can be enough to keep the elephant away.

BEAR:

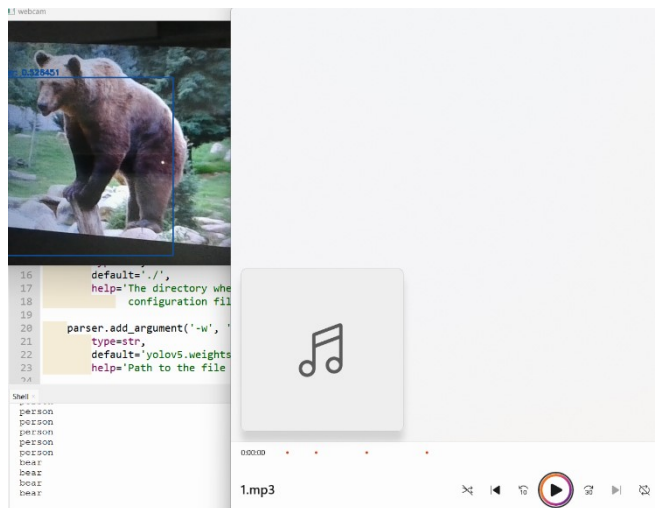


Fig. 6. Result when bear is detected.

When a bear is detected, the system will give alert to bear by playing loud sounds of bullets.

COW:

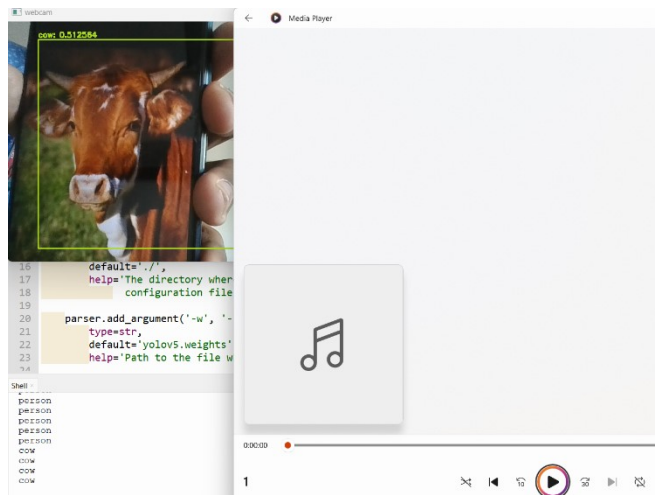


Fig. 7. Result when cow is detected.

When a cow is detected, the system will give alert to cow by playing loud sounds of bullets.

IX. FUTURE WORK

Improving Accuracy: The performance of the you only look once version5 (YOLO V5) algorithm can be enhance by utilizing more diverse and comprehensive datasets and fine-tuning the model using state-of-the-art techniques.

Integrating with Other Systems: A more complete solution for managing wild animals can be provided by integrating the Advanced Wild Animal Detection and Warning System with additional systems, such as global positioning system (GPS) tracking and wildlife monitoring systems.

Real-time Data Incorporation: To enhance the system's accuracy and give the alerts additional context, massive environmental data, such as weather and temperature, can be included.

Developing a Mobile Application: A mobile application can be developed to allow users to access the system and receive alerts on-the-go.

Improving Alert System: The alert system can be improved by incorporating more sophisticated methods for determining the potential danger of wild animals, such as incorporating animal behavior patterns and population density data.

Expanding the System to Other Regions: The system can be expanded to other regions with diverse wildlife populations to provide a more comprehensive solution for wild animal management.

Deployment in Developing Countries: The system can be deployed in developing countries to help manage and protect wildlife populations and improve the safety and well-being of communities and the environment.

X. CONCLUSION

The Advanced Wild Animal Detection and Alert System using you only look once version5 (YOLO V5) model is a cost-effective solution for detecting and tracking wild animals. The system leverages you only look once version5 (YOLO V5) object recognition model and tracking algorithms to provide real-time alerts in case of potential danger. The system can improve the safety and well-being of communities and the environment by providing timely and accurate information about wild animals. The system has the potential to make a significant contribution to wild animal management and can be deployed in various regions with

diverse wildlife populations. The future work for the system includes improving its accuracy, integrating it with other systems, incorporating real-time environmental data, developing a mobile application, and expanding the system to other regions.

XI. REFERENCE

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